

EXP 9,10,11

DSA0404-FDS

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EXP 9:

```
import pandas as pd

property_data = pd.DataFrame({
    "Property_ID": [101, 102, 103, 104, 105],
    "Location": ["Chennai", "Chennai", "Bangalore", "Hyderabad", "Bangalore"],
    "Bedrooms": [3, 5, 4, 6, 2],
    "Area_sqft": [1200, 2500, 1800, 3000, 1000],
    "Listing_Price": [5000000, 9000000, 7000000, 12000000, 4000000]
})

avg_price = property_data.groupby("Location")["Listing_Price"].mean()

count = property_data[property_data["Bedrooms"] > 4].shape[0]

largest_property = property_data.loc[property_data["Area_sqft"].idxmax()]

print("Average Listing Price:")

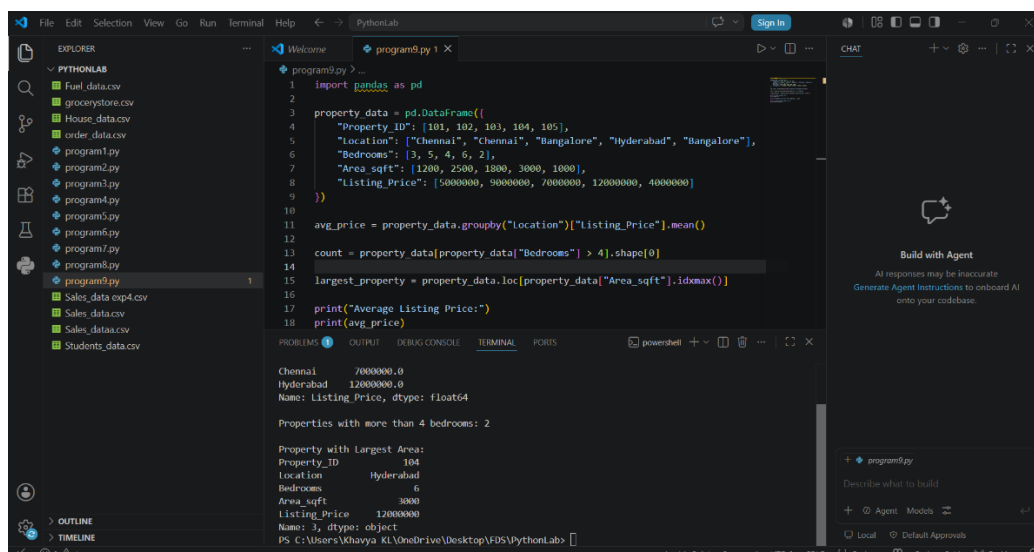
print(avg_price)

print("\nProperties with more than 4 bedrooms:", count)

print("\nProperty with Largest Area:")

print(largest_property)
```

OUTPUT:



The screenshot shows a Python IDE with the following code in `program9.py`:

```
1 import pandas as pd
2
3 property_data = pd.DataFrame({
4     "Property_ID": [101, 102, 103, 104, 105],
5     "Location": ["Chennai", "Chennai", "Bangalore", "Hyderabad", "Bangalore"],
6     "Bedrooms": [3, 5, 4, 6, 2],
7     "Area_sqft": [1200, 2500, 1800, 3000, 1000],
8     "Listing_Price": [5000000, 9000000, 7000000, 12000000, 4000000]
9 })
10
11 avg_price = property_data.groupby("Location")["Listing_Price"].mean()
12
13 count = property_data[property_data["Bedrooms"] > 4].shape[0]
14
15 largest_property = property_data.loc[property_data["Area_sqft"].idxmax()]
16
17 print("Average Listing Price:")
18 print(avg_price)
19
20 print("\nProperties with more than 4 bedrooms:", count)
21
22 print("\nProperty with Largest Area:")
23 print(largest_property)
```

The output in the terminal is as follows:

```
Chennai    7000000.0
Hyderabad  12000000.0
Name: Listing_Price, dtype: float64

Properties with more than 4 bedrooms: 2

Property with Largest Area:
Property_ID    104
Location      Hyderabad
Bedrooms        6
Area_sqft      3000
Listing_Price  12000000
Name: 3, dtype: object
PS C:\Users\Khaviya KL\OneDrive\Desktop\FDS\PythonLab>
```

EXP 10:

```
import matplotlib.pyplot as plt
```

```
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
```

```
sales = [120, 150, 180, 170, 200, 220]
```

```
plt.figure(1)
```

```
plt.plot(months, sales, marker='o')
```

```
plt.title("Monthly Sales - Line Plot")
```

```
plt.xlabel("Month")
```

```
plt.ylabel("Sales")
```

```
plt.figure(2)
```

```
plt.bar(months, sales)
```

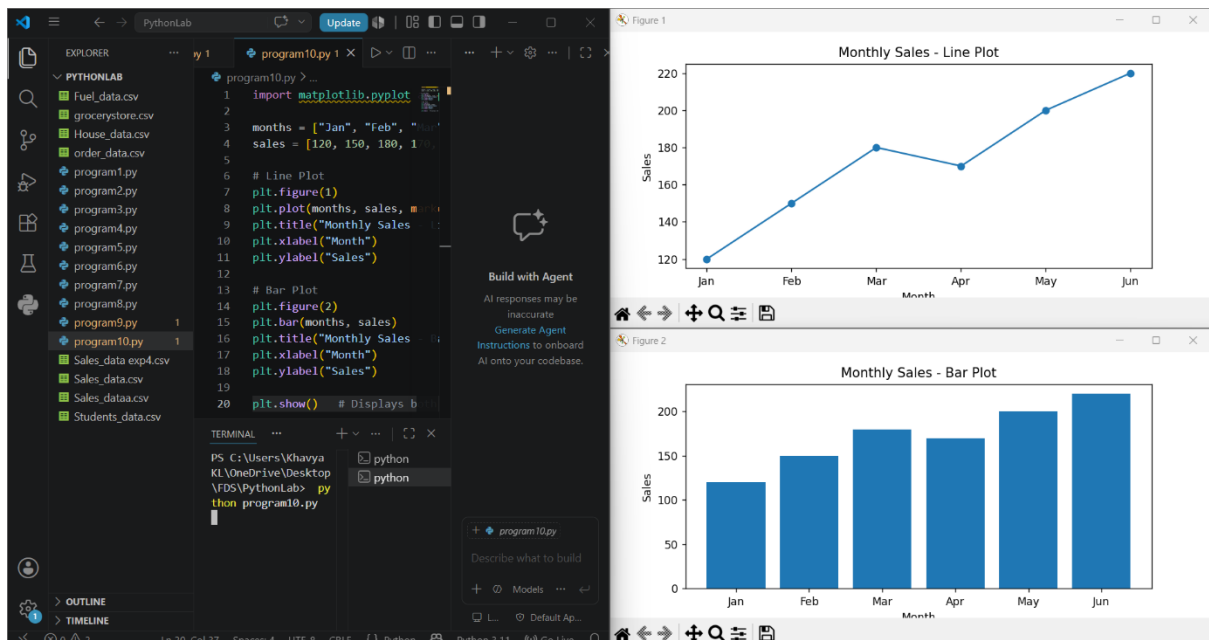
```
plt.title("Monthly Sales - Bar Plot")
```

```
plt.xlabel("Month")
```

```
plt.ylabel("Sales")
```

```
plt.show()
```

OUTPUT:



EXP 11:

```
import matplotlib.pyplot as plt
```

```
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
```

```
sales = [120, 150, 180, 170, 200, 220]
```

```
plt.figure(1)
```

```
plt.plot(months, sales, marker='o')
```

```
plt.title("Line Plot")
```

```
plt.figure(2)
```

```
plt.scatter(months, sales)
```

```
plt.title("Scatter Plot")
```

```
plt.figure(3)
```

```
plt.bar(months, sales)
```

```
plt.title("Bar Plot")
```

```
plt.show()
```

OUTPUT:

